



# IP ADDRESS SCHEME DESIGN

## & STATIC ROUTING WITH **RIP** CONFIGURATION

## Networking Concepts

Title of Assignment: IP Address Scheme Design and Static Routing

Module: CI5210

| DATE       | AUTHOR:                                                              | FOR:                                                                       |
|------------|----------------------------------------------------------------------|----------------------------------------------------------------------------|
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*This report includes the design, configuration, and testing of an IP address scheme with static routing and RIP implementation in a simulated network*

This report has been prepared by **Malik Omar (Student ID: K2335386)** as part of the **CI5210 Networking Concepts** coursework at Kingston University. The purpose of this report is to design and implement a **Variable Length Subnet Mask (VLSM)-based IP addressing scheme** for a small internetwork, configure **static routing**, and enable the **Routing Information Protocol (RIP v2)** to ensure network scalability and connectivity. This work reflects an in-depth understanding of IP subnetting, routing principles, and dynamic protocols, ensuring a robust and efficient network design.

# Introduction

The provided topology (Figure 1) consists of a Richmond site with two buildings, named Building A and Building B.

Figure 1:

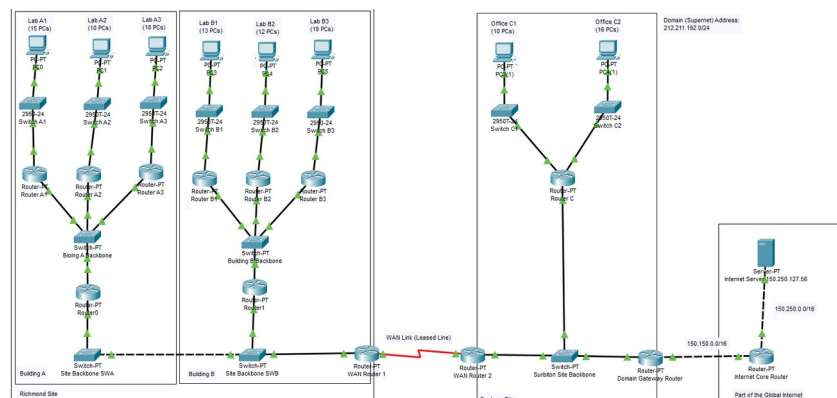


Figure 2: VLSM Calculations

- /27 Subnet (30 hosts per lab)
  - Formula:  $2^5 = 32$  total addresses
  - Usable IPs: 30 (32 - 2 for network and broadcast)
  - Example: 212.211.192.0/27 → Usable: 212.211.192.1 - 212.211.192.30
- /28 Subnet (14 hosts)
  - Formula:  $2^4 = 16$  total addresses
  - Usable IPs: 14 (16 - 2)
  - Example: 212.211.192.64/28 → Usable: 212.211.192.65 - 212.211.192.78
- /29 Subnet (6 hosts)
  - Formula:  $2^3 = 8$  total addresses
  - Usable IPs: 6 (8 - 2)
  - Example: 212.211.192.72/29 → Usable: 212.211.192.73 - 212.211.192.78
- /30 Subnet (2 hosts for WAN links)
  - Formula:  $2^2 = 4$  total addresses
  - Usable IPs: 2 (4 - 2)
  - Example: 212.211.192.224/30 → Usable: 212.211.192.225 - 212.211.192.226

The network consists of a Richmond site with two buildings, Building A and Building B. Each building has its own Site Backbone Switch and a Building Backbone Switch connected to a router. Inside the buildings, there are three labs, each equipped with a router and an access switch that connects the PCs. Buildings A and B are linked using their Site Backbone Switches, while the Building B Backbone Switch connects to a WAN Router, providing access to the Sunbury site.

At the Sunbury site, there are two offices, Office C1 and Office C2, each with its own access switch connected to a router. These routers link to the Sunbury Backbone Switch, which also connects to the Domain Gateway Router. The Domain Gateway Router then connects to the Internet Core Router, providing connectivity to an Internet Server.

## Subnet Summary (visualisation)

| SUBNET   | Network         | Subnet Mask     | Hosts | Device                 |
|----------|-----------------|-----------------|-------|------------------------|
| Subnet 1 | 212.211.192.0   | 255.255.255.224 | 30    | RouterA1, LAB A1       |
| Subnet 2 | 212.211.192.32  | 255.255.255.224 | 30    | RouterA3, LAB A3       |
| Subnet 3 | 212.211.192.224 | 255.255.255.252 | 2     | WANROUTER1, WANROUTER2 |

## VLSM Calculation

To efficiently allocate IP addresses, the 212.211.192.0/24 IP pool was divided into subnets of varying sizes based on device requirements: /27 Subnets: Allocated for larger segments, such as the labs, providing 30 usable IP addresses each (e.g., Subnet 1 and Subnet 2). /28 Subnets: Designed for smaller segments requiring up to 14 usable IP addresses. /29 Subnets: Used for point-to-point connections, offering 6 usable IP addresses. /30 Subnets: Allocated for WAN links, which require only 2 usable IP addresses (e.g., Subnet 3). This approach ensured optimal use of IP address space while meeting the requirements of the network topology. (The mathematical calculation is provided in [Figure 2](#))

## IP addressing Scheme

The IP pool 212.211.192.0/24 has been divided into smaller subnets, detailed in designated tables.

| IP NETWORK                    |                       | SUBNETTING DETAILS |                                     |
|-------------------------------|-----------------------|--------------------|-------------------------------------|
| SUBNET 1:<br>212.211.192.0/27 | DESCRIPTION           | DECIMAL            | BINARY (NETWORK: BLUE   HOST: RED)  |
|                               | Network:              | 212.211.192.0/27   | 11010100.11010011.11000000.00 00000 |
|                               | Subnet Mask           | 255.255.255.224    | 11111111.11111111.11111111.11 00000 |
|                               | HostMin:              | 212.211.192.1      | 11010100.11010011.11000000.00 00001 |
|                               | HostMax:              | 212.211.192.30     | 11010100.11010011.11000000.00 11110 |
|                               | Broadcast:            | 212.211.192.31     | 11010100.11010011.11000000.00 11111 |
|                               | Hosts/Net:            | 30                 |                                     |
|                               | CONFIGURATION DETAILS |                    |                                     |
|                               | DEVICE                | RouterA1           | PC LAB A1                           |
|                               | Interface             | FastEthernet0/0    | FastEthernet0                       |

| IP NETWORK                     |                       | SUBNETTING DETAILS |                                     |
|--------------------------------|-----------------------|--------------------|-------------------------------------|
|                                | Address               | 212.211.192.1      | 212.211.192.2                       |
| IP NETWORK                     |                       | SUBNETTING DETAILS |                                     |
| SUBNET 2:<br>212.211.192.32/27 | DESCRIPTION           | DECIMAL            | BINARY                              |
|                                | Network:              | 212.211.192.32/27  | 11010100.11010011.11000000.01 00000 |
|                                | Subnet Mask           | 255.255.255.224    | 11111111.11111111.11111111.11 00000 |
|                                | HostMin:              | 212.211.192.33     | 11010100.11010011.11000000.01 00001 |
|                                | HostMax:              | 212.211.192.62     | 11010100.11010011.11000000.01 11110 |
|                                | Broadcast:            | 212.211.192.63     | 11010100.11010011.11000000.01 11111 |
|                                | Hosts/Net:            | 30                 |                                     |
|                                | CONFIGURATION DETAILS |                    |                                     |
|                                | DEVICE                | RouterA3           | PC LAB A3                           |
|                                | Interface             | FastEthernet0/0    | FastEthernet0                       |
|                                | Address               | 212.211.192.33     | 212.211.192.34                      |
| IP NETWORK                     |                       | SUBNETTING DETAILS |                                     |
| SUBNET 3:<br>212.211.192.64/27 | DESCRIPTION           | DECIMAL            | BINARY                              |
|                                | Network:              | 212.211.192.64/27  | 11010100.11010011.11000000.10 00000 |
|                                | Subnet Mask           | 255.255.255.224    | 11111111.11111111.11111111.11 00000 |
|                                | HostMin:              | 212.211.192.65     | 11010100.11010011.11000000.10 00001 |
|                                | HostMax:              | 212.211.192.94     | 11010100.11010011.11000000.10 11110 |
|                                | Broadcast:            | 212.211.192.95     | 11010100.11010011.11000000.10 11111 |
|                                | Hosts/Net:            | 30                 |                                     |
|                                | CONFIGURATION DETAILS |                    |                                     |
|                                | DEVICE                | RouterB3           | PC LAB B3                           |
|                                | Interface             | FastEthernet0/0    | FastEthernet0                       |
|                                | Address               | 212.211.192.65     | 212.211.192.66                      |

| IP NETWORK                     |             | SUBNETTING DETAILS |                                     |
|--------------------------------|-------------|--------------------|-------------------------------------|
| SUBNET 4:<br>212.211.192.96/27 | DESCRIPTION | DECIMAL            | BINARY                              |
|                                | Network:    | 212.211.192.96/27  | 11010100.11010011.11000000.11 00000 |
|                                | Subnet Mask | 255.255.255.224    | 11111111.11111111.11111111.11 00000 |

| IP NETWORK                      |                       | SUBNETTING DETAILS |                                      |
|---------------------------------|-----------------------|--------------------|--------------------------------------|
|                                 | HostMin:              | 212.211.192.97     | 11010100.11010011.11000000.011 00001 |
|                                 | HostMax:              | 212.211.192.126    | 11010100.11010011.11000000.011 11110 |
|                                 | Broadcast:            | 212.211.192.127    | 11010100.11010011.11000000.011 11111 |
|                                 | Hosts/Net:            | 30                 |                                      |
|                                 | CONFIGURATION DETAILS |                    |                                      |
|                                 | DEVICE                | RouterC            | OFFICE C2                            |
|                                 | Interface             | FastEthernet0/0    | FastEthernet0                        |
|                                 | Address               | 212.211.192.97     | 212.211.192.98                       |
| IP NETWORK                      |                       | SUBNETTING DETAILS |                                      |
| SUBNET 5:<br>212.211.192.128/28 | DESCRIPTION           | DECIMAL            | BINARY                               |
|                                 | Network:              | 212.211.192.128/28 | 11010100.11010011.11000000.1000 0000 |
|                                 | Subnet Mask           | 255.255.255.240    | 11111111.11111111.11111111.111 0000  |
|                                 | HostMin:              | 212.211.192.129    | 11010100.11010011.11000000.1000 0001 |
|                                 | HostMax:              | 212.211.192.142    | 11010100.11010011.11000000.1000 1110 |
|                                 | Broadcast:            | 212.211.192.143    | 11010100.11010011.11000000.1000 1111 |
|                                 | Hosts/Net:            | 14                 |                                      |
|                                 | CONFIGURATION DETAILS |                    |                                      |
|                                 | DEVICE                | RouterB1           | PC LAB B1                            |
|                                 | Interface             | FastEthernet0/0    | FastEthernet0                        |
|                                 | Address               | 212.211.192.129    | 212.211.192.130                      |
| IP NETWORK                      |                       | SUBNETTING DETAILS |                                      |
| SUBNET 6:<br>212.211.192.144/28 | DESCRIPTION           | DECIMAL            | BINARY                               |
|                                 | Network:              | 212.211.192.144/28 | 11010100.11010011.11000000.1001 0000 |
|                                 | Subnet Mask           | 255.255.255.240    | 11111111.11111111.11111111.111 0000  |
|                                 | HostMin:              | 212.211.192.145    | 11010100.11010011.11000000.1001 0001 |
|                                 | HostMax:              | 212.211.192.158    | 11010100.11010011.11000000.1001 1110 |
|                                 | Broadcast:            | 212.211.192.159    | 11010100.11010011.11000000.1001 1111 |
|                                 | Hosts/Net:            | 14                 |                                      |
|                                 | CONFIGURATION DETAILS |                    |                                      |
|                                 | DEVICE                | RouterB2           | PC LAB B2                            |
|                                 | Interface             | FastEthernet0/0    | FastEthernet0                        |

| IP NETWORK |         | SUBNETTING DETAILS |                 |
|------------|---------|--------------------|-----------------|
|            | Address | 212.211.192.145    | 212.211.192.146 |

| IP NETWORK                      |                       | SUBNETTING DETAILS |                                      |
|---------------------------------|-----------------------|--------------------|--------------------------------------|
| SUBNET 7:<br>212.211.192.160/27 | DESCRIPTION           | DECIMAL            | BINARY                               |
|                                 | Network:              | 212.211.192.160/28 | 11010100.11010011.11000000.1010 0000 |
|                                 | Subnet Mask           | 255.255.255.240    | 11111111.11111111.11111111.111 0000  |
|                                 | HostMin:              | 212.211.192.161    | 11010100.11010011.11000000.1010 0001 |
|                                 | HostMax:              | 212.211.192.174    | 11010100.11010011.11000000.1010 1110 |
|                                 | Broadcast:            | 212.211.192.175    | 11010100.11010011.11000000.1010 1111 |
|                                 | Hosts/Net:            | 14                 |                                      |
|                                 | CONFIGURATION DETAILS |                    |                                      |
|                                 | DEVICE                | RouterA2           | PC LAB A2                            |
|                                 | Interface             | FastEthernet0/0    | FastEthernet0                        |
|                                 | Address               | 212.211.192.161    | 212.211.192.162                      |

| IP NETWORK                      |                       | SUBNETTING DETAILS |                                      |
|---------------------------------|-----------------------|--------------------|--------------------------------------|
| SUBNET 8:<br>212.211.192.176/28 | DESCRIPTION           | DECIMAL            | BINARY                               |
|                                 | Network:              | 212.211.192.176/28 | 11010100.11010011.11000000.1011 0000 |
|                                 | Subnet Mask           | 255.255.255.240    | 11111111.11111111.11111111.111 0000  |
|                                 | HostMin:              | 212.211.192.177    | 11010100.11010011.11000000.1011 0001 |
|                                 | HostMax:              | 212.211.192.190    | 11010100.11010011.11000000.1011 1110 |
|                                 | Broadcast:            | 212.211.192.191    | 11010100.11010011.11000000.1011 1111 |
|                                 | Hosts/Net:            | 14                 |                                      |
|                                 | CONFIGURATION DETAILS |                    |                                      |
|                                 | DEVICE                | RouterC            | OFFICE C2                            |
|                                 | Interface             | Ethernet6/0        | FastEthernet0                        |
|                                 | Address               | 212.211.192.177    | 212.211.192.178                      |

| IP NETWORK                      |             | SUBNETTING DETAILS |                                     |
|---------------------------------|-------------|--------------------|-------------------------------------|
| SUBNET 9:<br>212.211.192.192/29 | DESCRIPTION | DECIMAL            | BINARY                              |
|                                 | Network:    | 212.211.192.192/29 | 11010100.11010011.11000000.1000 000 |
|                                 | Subnet Mask | 255.255.255.248    | 11111111.11111111.11111111.1111 000 |

| IP NETWORK |                       | SUBNETTING DETAILS |                                      |
|------------|-----------------------|--------------------|--------------------------------------|
|            | HostMin:              | 212.211.192.193    | 11010100.11010011.11000000.11000 001 |
|            | HostMax:              | 212.211.192.198    | 11010100.11010011.11000000.11000 110 |
|            | Broadcast:            | 212.211.192.199    | 11010100.11010011.11000000.11000 111 |
|            | Hosts/Net:            | 6                  |                                      |
|            | CONFIGURATION DETAILS |                    |                                      |
|            | DEVICE                | Router0            | RouterA1                             |
|            | Interface             | FastEthernet0/0    | FastEthernet1/0                      |
|            | Address               | 212.211.192.193    | 212.211.192.194                      |
|            | DEVICE                | RouterA2           | RouterA3                             |
|            | Interface             | FastEthernet1/0    | FastEthernet1/0                      |
|            | Address               | 212.211.192.195    | 212.211.192.196                      |

| IP NETWORK                       |                       | SUBNETTING DETAILS |                                      |
|----------------------------------|-----------------------|--------------------|--------------------------------------|
| SUBNET 10:<br>212.211.192.200/29 | DESCRIPTION           | DECIMAL            | BINARY                               |
|                                  | Network:              | 212.211.192.200/29 | 11010100.11010011.11000000.11001 000 |
|                                  | Subnet Mask           | 255.255.255.248    | 11111111.11111111.11111111.1111 000  |
|                                  | HostMin:              | 212.211.192.201    | 11010100.11010011.11000000.11001 001 |
|                                  | HostMax:              | 212.211.192.206    | 11010100.11010011.11000000.11001 110 |
|                                  | Broadcast:            | 212.211.192.207    | 11010100.11010011.11000000.11001 111 |
|                                  | Hosts/Net:            | 6                  |                                      |
|                                  | CONFIGURATION DETAILS |                    |                                      |
|                                  | DEVICE                | Router1            | RouterB1                             |
|                                  | Interface             | FastEthernet0/0    | FastEthernet1/0                      |
|                                  | Address               | 212.211.192.201    | 212.211.192.202                      |
|                                  | DEVICE                | RouterB2           | RouterB3                             |
|                                  | Interface             | FastEthernet1/0    | FastEthernet1/0                      |
|                                  | Address               | 212.211.192.203    | 212.211.192.204                      |
| IP NETWORK                       |                       | SUBNETTING DETAILS |                                      |
| SUBNET 11:<br>212.211.192.208/29 | DESCRIPTION           | DECIMAL            | BINARY                               |
|                                  | Network:              | 212.211.192.208/29 | 11010100.11010011.11000000.11010 000 |
|                                  | Subnet Mask           | 255.255.255.248    | 11111111.11111111.11111111.1111 000  |

| IP NETWORK |                       | SUBNETTING DETAILS |                                      |
|------------|-----------------------|--------------------|--------------------------------------|
|            | HostMin:              | 212.211.192.209    | 11010100.11010011.11000000.11010 001 |
|            | HostMax:              | 212.211.192.214    | 11010100.11010011.11000000.11010 110 |
|            | Broadcast:            | 212.211.192.215    | 11010100.11010011.11000000.11010 111 |
|            | Hosts/Net:            | 6                  |                                      |
|            | CONFIGURATION DETAILS |                    |                                      |
|            | DEVICE                | WANROUTER1         | Router0                              |
|            | Interface             | FastEthernet0/0    | FastEthernet1/0                      |
|            | Address               | 212.211.192.209    | 212.211.192.210                      |
|            | DEVICE                | Router1            |                                      |
|            | Interface             | FastEthernet1/0    |                                      |
|            | Address               | 212.211.192.211    |                                      |

| IP NETWORK                       |                       | SUBNETTING DETAILS  |                                      |
|----------------------------------|-----------------------|---------------------|--------------------------------------|
| SUBNET 12:<br>212.211.192.216/29 | DESCRIPTION           | DECIMAL             | BINARY                               |
|                                  | Network:              | 212.211.192.216/29  | 11010100.11010011.11000000.11011 000 |
|                                  | Subnet Mask           | 255.255.255.248     | 11111111.11111111.11111111.1111 000  |
|                                  | HostMin:              | 212.211.192.217     | 11010100.11010011.11000000.11011 001 |
|                                  | HostMax:              | 212.211.192.222     | 11010100.11010011.11000000.11011 110 |
|                                  | Broadcast:            | 212.211.192.223     | 11010100.11010011.11000000.11011 111 |
|                                  | Hosts/Net:            | 6                   |                                      |
|                                  | CONFIGURATION DETAILS |                     |                                      |
|                                  | DEVICE                | WANROUTER2          | RouterC                              |
|                                  | Interface             | FastEthernet0/0     | FastEthernet1/0                      |
|                                  | Address               | 212.211.192.219     | 212.211.192.218                      |
|                                  | DEVICE                | DomainGatewayRouter |                                      |
|                                  | Interface             | FastEthernet0/0     |                                      |
|                                  | Address               | 212.211.192.217     |                                      |
| IP NETWORK                       |                       | SUBNETTING DETAILS  |                                      |
| SUBNET 13:                       | DESCRIPTION           | DECIMAL             | BINARY                               |



| IP NETWORK         |                       | SUBNETTING DETAILS |                                             |
|--------------------|-----------------------|--------------------|---------------------------------------------|
| 212.211.192.224/30 | Network:              | 212.211.192.224/30 | 11010100.11010011.11000000.11000 00         |
|                    | Subnet Mask           | 255.255.255.252    | 11111111.11111111.11111111.11111111.1111 00 |
|                    | HostMin:              | 212.211.192.225    | 11010100.11010011.11000000.11000 01         |
|                    | HostMax:              | 212.211.192.226    | 11010100.11010011.11000000.11000 10         |
|                    | Broadcast:            | 212.211.192.227    | 11010100.11010011.11000000.11000 11         |
|                    | Hosts/Net:            | 2                  |                                             |
|                    | CONFIGURATION DETAILS |                    |                                             |
|                    | DEVICE                | WANROUTER1         | WANROUTER2                                  |
|                    | Interface             | Serial2/0          | Serial2/0                                   |
|                    | Address               | 212.211.192.226    | 212.211.192.225                             |

## Routing Configuration

### WHAT IS STATIC ROUTING?

Static routing is a method where routes are manually configured on routers to direct traffic to specific networks. Unlike dynamic routing, it provides fixed, predefined paths for data to travel.

### WHY STATIC ROUTES WERE USED

1. Precise and Reliable: Ensures predictable and accurate routing.
2. Simplicity: Ideal for smaller, stable networks with minimal changes.
3. Control: Allows fine-grained control over internal traffic flow.

### STEPS TAKEN FOR STATIC ROUTING

1. Internal Routes: Configured static routes for each subnet to direct packets based on destination IP.
  - Example: `ip route <destination-network> <subnet-mask> <next-hop IP>`
2. Default Route: Set a default route to forward traffic for unknown networks to a predefined gateway.

- Example: `ip route 0.0.0.0 0.0.0.0 <gateway IP>`

### WHAT IS RIP (DYNAMIC ROUTING)?

RIP (Routing Information Protocol) is a dynamic routing protocol that allows routers to automatically update their routing tables by exchanging information with neighbouring routers. It uses hop count as a metric to determine the best path for data to travel.

### WHY RIP WAS USED

1. Automatic Updates: Routes are updated dynamically, eliminating the need for manual configuration when network changes occur (e.g., link failures, new routes).
2. Scalability: Ideal for growing networks with multiple routers and subnets, as it adapts automatically to changes in topology.
3. Simplicity: RIP is easy to configure and maintain, making it suitable for mid-sized networks.

### WHAT ARE ROUTING TABLES?

Routing tables are used by routers to determine the best path for forwarding packets across the network. Static routing was implemented for this network to ensure precise control over routing decisions.

The table “WAN Router 2” demonstrate the difference between static routing and dynamic routing using RIP v2. Static routes were manually configured for precise control, while RIP v2 enabled dynamic route learning to automate updates.

### WHY TWO TABLES?

The configurations for WAN Router 2 are shown in two separate screenshots because static routing and RIP routing cannot coexist for the same destinations in a routing table.

Static routes have a lower administrative distance (1) compared to RIP routes (120), so the router will always pick static routes if they are configured for the same network.

To demonstrate both routing methods:

Static Configuration: Static routes were manually configured and displayed in the routing table.

RIP Configuration: Static routes were removed to allow RIP routes (R) to appear in the table dynamically.

| WAN ROUTER 2         |
|----------------------|
| STATIC CONFIGURATION |

## WAN ROUTER 2

```

WANROUTER2#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 212.211.192.217 to network 0.0.0.0

    212.211.192.0/24 is variably subnetted, 10 subnets, 4 masks
S       212.211.192.0/27 [1/0] via 212.211.192.226
S       212.211.192.32/27 [1/0] via 212.211.192.226
S       212.211.192.64/27 [1/0] via 212.211.192.226
S       212.211.192.96/27 [1/0] via 212.211.192.218
S       212.211.192.128/28 [1/0] via 212.211.192.226
S       212.211.192.144/28 [1/0] via 212.211.192.226
S       212.211.192.160/28 [1/0] via 212.211.192.226
S       212.211.192.176/28 [1/0] via 212.211.192.218
C       212.211.192.216/29 is directly connected, FastEthernet0/0
C       212.211.192.224/30 is directly connected, Serial2/0
S*    0.0.0.0/0 [1/0] via 212.211.192.217

```

## RIP CONFIGURATION:

```

WANROUTER2#sho ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

R       150.150.0.0/16 [120/1] via 212.211.192.217, 00:00:19, FastEthernet0/0
R       150.250.0.0/16 [120/2] via 212.211.192.217, 00:00:19, FastEthernet0/0
    212.211.192.0/24 is variably subnetted, 13 subnets, 4 masks
R       212.211.192.0/27 [120/3] via 212.211.192.226, 00:00:19, Serial2/0
R       212.211.192.32/27 [120/3] via 212.211.192.226, 00:00:19, Serial2/0
R       212.211.192.64/27 [120/3] via 212.211.192.226, 00:00:19, Serial2/0
R       212.211.192.96/27 [120/1] via 212.211.192.218, 00:00:18, FastEthernet0/0
R       212.211.192.128/28 [120/3] via 212.211.192.226, 00:00:19, Serial2/0
R       212.211.192.144/28 [120/3] via 212.211.192.226, 00:00:19, Serial2/0
R       212.211.192.160/28 [120/3] via 212.211.192.226, 00:00:19, Serial2/0
R       212.211.192.176/28 [120/1] via 212.211.192.218, 00:00:18, FastEthernet0/0
R       212.211.192.192/29 [120/2] via 212.211.192.226, 00:00:19, Serial2/0
R       212.211.192.200/29 [120/2] via 212.211.192.226, 00:00:19, Serial2/0
R       212.211.192.208/29 [120/1] via 212.211.192.226, 00:00:19, Serial2/0
C       212.211.192.216/29 is directly connected, FastEthernet0/0
C       212.211.192.224/30 is directly connected, Serial2/0

```

## STATIC + RIP + LAST RESORT

## WAN ROUTER 2

WANROUTER2>sh ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP  
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
 E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP  
 i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area  
 \* - candidate default, U - per-user static route, o - ODR  
 P - periodic downloaded static route

Gateway of last resort is 212.211.192.217 to network 0.0.0.0

```
R    150.150.0.0/16 [120/1] via 212.211.192.217, 00:00:09, FastEthernet0/0
R    150.250.0.0/16 [120/2] via 212.211.192.217, 00:00:09, FastEthernet0/0
    212.211.192.0/24 is variably subnetted, 13 subnets, 4 masks
S     212.211.192.0/27 [1/0] via 212.211.192.226
S     212.211.192.32/27 [1/0] via 212.211.192.226
S     212.211.192.64/27 [1/0] via 212.211.192.226
S     212.211.192.96/27 [1/0] via 212.211.192.218
S     212.211.192.128/28 [1/0] via 212.211.192.226
S     212.211.192.144/28 [1/0] via 212.211.192.226
S     212.211.192.160/28 [1/0] via 212.211.192.226
S     212.211.192.176/28 [1/0] via 212.211.192.218
R     212.211.192.192/29 [120/2] via 212.211.192.226, 00:00:13, Serial2/0
R     212.211.192.200/29 [120/2] via 212.211.192.226, 00:00:13, Serial2/0
R     212.211.192.208/29 [120/1] via 212.211.192.226, 00:00:13, Serial2/0
C     212.211.192.216/29 is directly connected, FastEthernet0/0
C     212.211.192.224/30 is directly connected, Serial2/0
S*   0.0.0.0/0 [1/0] via 212.211.192.217
```

## ROUTER A1

### STATIC CONFIGURATION

## ROUTER 1

### STATIC CONFIGURATION

```
Router1#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 212.211.192.209 to network 0.0.0.0

    212.211.192.0/24 is variably subnetted, 5 subnets, 3 masks
S       212.211.192.64/27 [1/0] via 212.211.192.204
S       212.211.192.128/28 [1/0] via 212.211.192.202
S       212.211.192.144/28 [1/0] via 212.211.192.203
C       212.211.192.200/29 is directly connected, FastEthernet0/0
C       212.211.192.208/29 is directly connected, FastEthernet1/0
S*     0.0.0.0/0 [1/0] via 212.211.192.209
```

### RIP CONFIGURATION:



# ROUTER 1

```
Router1#sho ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

R      150.150.0.0/16 [120/3] via 212.211.192.209, 00:00:02, FastEthernet1/0
R      150.250.0.0/16 [120/4] via 212.211.192.209, 00:00:02, FastEthernet1/0
      212.211.192.0/24 is variably subnetted, 13 subnets, 4 masks
R      212.211.192.0/27 [120/2] via 212.211.192.210, 00:00:07, FastEthernet1/0
R      212.211.192.32/27 [120/2] via 212.211.192.210, 00:00:07, FastEthernet1/0
R      212.211.192.64/27 [120/1] via 212.211.192.204, 00:00:20, FastEthernet0/0
R      212.211.192.96/27 [120/3] via 212.211.192.209, 00:00:02, FastEthernet1/0
R      212.211.192.128/28 [120/1] via 212.211.192.202, 00:00:12, FastEthernet0/0
R      212.211.192.144/28 [120/1] via 212.211.192.203, 00:00:28, FastEthernet0/0
R      212.211.192.160/28 [120/2] via 212.211.192.210, 00:00:07, FastEthernet1/0
R      212.211.192.176/28 [120/3] via 212.211.192.209, 00:00:02, FastEthernet1/0
R      212.211.192.192/29 [120/1] via 212.211.192.210, 00:00:07, FastEthernet1/0
C      212.211.192.200/29 is directly connected, FastEthernet0/0
C      212.211.192.208/29 is directly connected, FastEthernet1/0
R      212.211.192.216/29 [120/2] via 212.211.192.209, 00:00:02, FastEthernet1/0
R      212.211.192.224/30 [120/1] via 212.211.192.209, 00:00:02, FastEthernet1/0
```

STATIC + RIP + LAST RESORT

# ROUTER 1

Router1>sh ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP  
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
 E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP  
 i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area  
 \* - candidate default, U - per-user static route, o - ODR  
 P - periodic downloaded static route

Gateway of last resort is 212.211.192.209 to network 0.0.0.0

```
R    150.150.0.0/16 [120/3] via 212.211.192.209, 00:00:07, FastEthernet1/0
R    150.250.0.0/16 [120/4] via 212.211.192.209, 00:00:07, FastEthernet1/0
    212.211.192.0/24 is variably subnetted, 13 subnets, 4 masks
R      212.211.192.0/27 [120/2] via 212.211.192.210, 00:00:18, FastEthernet1/0
R      212.211.192.32/27 [120/2] via 212.211.192.210, 00:00:18, FastEthernet1/0
S      212.211.192.64/27 [1/0] via 212.211.192.204
R      212.211.192.96/27 [120/3] via 212.211.192.209, 00:00:07, FastEthernet1/0
S      212.211.192.128/28 [1/0] via 212.211.192.202
S      212.211.192.144/28 [1/0] via 212.211.192.203
R      212.211.192.160/28 [120/2] via 212.211.192.210, 00:00:18, FastEthernet1/0
R      212.211.192.176/28 [120/3] via 212.211.192.209, 00:00:07, FastEthernet1/0
R      212.211.192.192/29 [120/1] via 212.211.192.210, 00:00:18, FastEthernet1/0
C      212.211.192.200/29 is directly connected, FastEthernet0/0
C      212.211.192.208/29 is directly connected, FastEthernet1/0
R      212.211.192.216/29 [120/2] via 212.211.192.209, 00:00:07, FastEthernet1/0
R      212.211.192.224/30 [120/1] via 212.211.192.209, 00:00:07, FastEthernet1/0
S*    0.0.0.0/0 [1/0] via 212.211.192.209
```

DOMAIN GATEWAY ROUTER

STATIC CONFIGURATION

## Testing and Evaluation

---

FROM PC LAB A1 (PC0)

PING OFFICE C1:

 PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 212.211.192.98

Pinging 212.211.192.98 with 32 bytes of data:

Reply from 212.211.192.98: bytes=32 time=1ms TTL=122
Reply from 212.211.192.98: bytes=32 time=2ms TTL=122
Reply from 212.211.192.98: bytes=32 time=1ms TTL=122
Reply from 212.211.192.98: bytes=32 time=3ms TTL=122

Ping statistics for 212.211.192.98:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 1ms
```

PING INTERNET SERVER:



 PC0
 

Physical   Config   **Desktop**   Programming   Attributes

**Command Prompt**

```

C:\>ping 150.250.127.56

Pinging 150.250.127.56 with 32 bytes of data:

Reply from 150.250.127.56: bytes=32 time=1ms TTL=122
Reply from 150.250.127.56: bytes=32 time=1ms TTL=122
Reply from 150.250.127.56: bytes=32 time=28ms TTL=122
Reply from 150.250.127.56: bytes=32 time=1ms TTL=122

Ping statistics for 150.250.127.56:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 28ms, Average = 7ms

```

TRACEROUTE OFFICE C1:

 PC0
 

Physical   Config   **Desktop**   Programming   Attributes

**Command Prompt**

```

C:\>tracert 212.211.192.98

Tracing route to 212.211.192.98 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    212.211.192.1
  2  0 ms    0 ms    0 ms    212.211.192.193
  3  0 ms    0 ms    0 ms    212.211.192.209
  4  1 ms    1 ms    1 ms    212.211.192.225
  5  1 ms    1 ms    1 ms    212.211.192.218
  6  1 ms    0 ms    0 ms    212.211.192.98

Trace complete.

```

TRACEROUTE INTERNET SERVER:

FROM PC LAB A1 (PC0)

PC0

Physical Config **Desktop** Programming Attributes

Command Prompt

```
C:\>tracert 150.250.127.56

Tracing route to 150.250.127.56 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    212.211.192.1
  2  0 ms    0 ms    0 ms    212.211.192.193
  3  0 ms    0 ms    0 ms    212.211.192.209
  4  0 ms    1 ms    1 ms    212.211.192.225
  5  1 ms    1 ms    0 ms    212.211.192.217
  6  1 ms    0 ms    0 ms    150.150.0.2
  7  0 ms    1 ms    0 ms    150.250.127.56

Trace complete.
```

WEBPAGE ON INTERNET SERVER:



FROM OFFICE C2 (PC4(1))

PING LAB B2:

 PC4(1)

Physical

Config

Desktop

Programming

Attributes

## Command Prompt

```
C:\>ping 212.211.192.146
```

```
Pinging 212.211.192.146 with 32 bytes of data:
```

```
Reply from 212.211.192.146: bytes=32 time=1ms TTL=123
Reply from 212.211.192.146: bytes=32 time=1ms TTL=123
Reply from 212.211.192.146: bytes=32 time=1ms TTL=123
Reply from 212.211.192.146: bytes=32 time=10ms TTL=123
```

```
Ping statistics for 212.211.192.146:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 10ms, Average = 3ms
```

PING INTERNET SERVER:

 PC4(1)

Physical

Config

Desktop

Programming

Attributes

## Command Prompt

```
C:\>ping 150.250.127.56
```

```
Pinging 150.250.127.56 with 32 bytes of data:
```

```
Reply from 150.250.127.56: bytes=32 time<1ms TTL=125
Reply from 150.250.127.56: bytes=32 time<1ms TTL=125
Reply from 150.250.127.56: bytes=32 time<1ms TTL=125
Reply from 150.250.127.56: bytes=32 time<1ms TTL=125
```

```
Ping statistics for 150.250.127.56:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

TRACEROUTE LAB B2:

PC4(1)

Physical

Config

Desktop

Programming

Attributes

## Command Prompt

```
C:\>tracert 212.211.192.146
```

```
Tracing route to 212.211.192.146 over a maximum of 30 hops:
```

|   |      |      |      |                 |
|---|------|------|------|-----------------|
| 1 | 0 ms | 0 ms | 0 ms | 212.211.192.97  |
| 2 | 0 ms | 0 ms | 0 ms | 212.211.192.217 |
| 3 | 0 ms | 0 ms | 0 ms | 212.211.192.219 |
| 4 | 0 ms | 1 ms | 1 ms | 212.211.192.226 |
| 5 | 1 ms | 1 ms | 0 ms | 212.211.192.211 |
| 6 | 2 ms | 1 ms | 1 ms | 212.211.192.203 |
| 7 | 0 ms | 0 ms | 0 ms | 212.211.192.146 |

```
Trace complete.
```

TRACEROUTE INTERNET SERVER:

PC4(1)

Physical

Config

Desktop

Programming

Attributes

## Command Prompt

```
C:\>tracert 150.250.127.56
```

```
Tracing route to 150.250.127.56 over a maximum of 30 hops:
```

|   |      |      |      |                 |
|---|------|------|------|-----------------|
| 1 | 0 ms | 0 ms | 0 ms | 212.211.192.97  |
| 2 | 0 ms | 0 ms | 0 ms | 212.211.192.217 |
| 3 | 0 ms | 0 ms | 0 ms | 150.150.0.2     |
| 4 | 0 ms | 0 ms | 0 ms | 150.250.127.56  |

```
Trace complete.
```

## Conclusion

---

The objective of this coursework was to design and configure a working network from a provided Packet Tracer (PKT) file that had no routing configuration at the beginning. The network topology provided was a combination of some subnets spread out over different devices, such as routers, PCs and switches. The challenge was to come up with an efficient IP addressing scheme, set up routing from scratch, and get all devices within the network to communicate with each other, and with external destinations.

First, to optimally divide the available IP address space amongst subnets of varying sizes, Variable Length Subnet Masking (VLSM) was applied. Addresses were assigned to each subnet in a careful way to minimize waste and to meet the connectivity requirements of the network.

Static routes were configured manually on all routers following the IP addressing scheme. Here, routes were set up for each subnet and a default route (last resort) was implemented to send traffic to external networks if there was no specific route. Precise control of packet forwarding was guaranteed by static routing, but manual configuration was shown to be limited by the size of the network.

In order to overcome these limitations and show scalability, Routing Information Protocol (RIP) was implemented as a dynamic routing protocol. RIP made it possible for routers to automatically exchange and update routing information, thus reducing administrative overhead and facilitating adaptation to network changes.

In summary, the project consisted of setting up the network from scratch, IP addressing, static routing, default routes, and RIP configuration. It was an educational project which taught a lot about the fundamentals of network design; subnetting, routing protocols, troubleshooting techniques. It was an excellent hands on experience to understand how different routing methods operate, their pros and cons, and how critical the configuration is to make a complete and efficient network.

### KEY TAKEAWAYS

- **Subnetting with VLSM:** Efficiently dividing IP address space based on host requirements.
- **Static Routing:** Precise control but complex for larger networks.
- **Dynamic Routing:** RIP's automation simplifies management and enhances scalability.
- **Connectivity Testing:** Using tools like ping and tracert to verify network performance.

Overall, this project provided hands-on experience with IP addressing, static routing, and dynamic protocols, enhancing my ability to design and troubleshoot functional networks.